Visual Counterfactual Explanations and Beyond: A Systematic Review

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Visual counterfactuals are increasingly used for learning, auditing, benchmarking, and inference-time mitigation. However, their representations and validation methods remain disjointed, complicating the process of determining when a class-flipping artifact substantiates an explanatory, domain-level, or downstream assertion. We conduct a systematic review that critically analyzes 95 primary studies. We define the concept of a visual counterfactual operationally via its factual-counterfactual relationship. We also distinguish the artifact from the technical role it fulfills and organize the literature through a six-dimensional taxonomy. We further analyze the evolution of methodologies and application patterns and investigate evaluation through claim-metric alignment. Our findings indicate that successive methodological families have enhanced realism, controllability, and scope, while also introducing dependencies on factual examples, attribution estimators, learned generative priors, semantic evaluators, or foundational-model components. Across these frameworks, measures such as target change, perceptual realism, and human preferences offer various forms of evidence; however, none independently confirm semantic validity, domain feasibility, or downstream utility. Consequently, we call for claim-centered evaluation, in which the counterfactuals, permissible interventions, and intended role are explicitly articulated and tested with suitably independent evidence. This perspective facilitates meaningful comparison without treating heterogeneous visual counterfactuals as interchangeable explanations or reducing their evaluation to a universal score.

CCS Concepts: • **General and reference** → **Surveys and overviews**; • **Computing methodologies** → **Reasoning about belief and knowledge**; **Image representations**.

Additional Key Words and Phrases: Counterfactual learning, Explainable AI, Generative models, Multimodal reasoning, Visual counterfactual explanations.

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1 Introduction

Counterfactual explanation transforms prediction analysis into a controlled comparison, such as what must be different for a target model f to generate outcome $y^\star$ instead of y? This formulation embodies the contrastive nature of explanatory inquiries while establishing an operational framework for developing and evaluating alternatives [Guidotti 2024; Miller 2021; Verma et al. 2024]. In computer vision, however, the answer is rarely reducible to a minimal input perturbation [Goyal et al. 2019]. Visual features are high-dimensional and semantically entangled; thus, effective model modifications may be imperceptible or meaningless, whereas photorealistic edits may inadvertently alter unintended content or even depend on correlations that are not pertinent to the target model. Therefore, a plausible visual counterfactual must explicitly specify not only the desired outcome (or *foil*) but also the relevant contrast, permissible changes, and the properties that should remain invariant.

Visual counterfactuals represent this relationship through various forms. They might replace a specific discriminative region [Goyal et al. 2019], pinpoint evidence in relation to a counter-class [Wang and Vasconcelos 2020], alter an image using a learned generative prior [Augustin et al. 2022], change a visual state associated with an alternative action [Huber et al. 2023], or create a controlled pair of factual–counterfactual pair for evaluation [Jiang et al. 2026]. Each approach involves different evidential commitments for validation of the generated output (or *artifacts*). A contrastive map may show where evidence relative to the model is located without providing a feasible alternative, whereas a generated image might offer an explicit counterfactual but transfers part of its validity to the generator, the semantic evaluator, and the preservation constraints. Considering these artifacts as interchangeable could obscure what has been directly modified and the evidence their explanation can support.

The technical roles of visual counterfactuals have also expanded beyond post-hoc explanation. Factual–counterfactual relations are now employed to enhance training data and modify predictive behavior [Li et al. 2024c; Singla et al. 2023b], support visually grounded learning and evaluation [Yu and Li 2024; Zhang et al. 2024a], and intervene in multimodal training or inference pipelines [Howard et al. 2024; Liang et al. 2026]. This expansion creates a significant distinction between the output (or artifact) produced by a method and the purpose for which it is utilized. An image that improves downstream performance may serve as valuable training data without accurately explaining an individual prediction; similarly, an artifact that reveals model sensitivity may support auditing without being semantically feasible within the application domain. Consequently, evidence of success cannot be automatically transferred across explanation, learning, auditing, benchmarking, mitigation, and repair.

Existing surveys have established general definitions, desiderata, algorithmic approaches, and benchmarking methods for counterfactual explanation and algorithmic recourse [Guidotti 2024; Verma et al. 2024]. Nevertheless, the visual literature's depiction of intervention, dependencies related to generation and model access, application-specific invariants, emerging roles at the system level, and the limitations of validation metrics are less thoroughly integrated. Addressing these challenges requires a multidimensional framework that specifies what is changed, how the counterfactual is constructed, the necessary information and auxiliary models, the reasoning behind using the counterfactual, and the claims supported by its evaluation.

We therefore conduct a systematic review with critical technical analysis. We define a *visual counterfactual operationally as an implemented comparison anchored in a factual case and a specified foil.* It involves changing, controlling, or recombining one or more visually grounded features to alter a model-relevant outcome. This qualification depends on the relationship between the factual and counterfactual examples, not necessarily on a specific output format or architecture. This definition encompasses post-hoc explanations and counterfactual-guided technical uses without assuming that every artifact qualifies as an explanation. It excludes, among other boundary cases, adversarial perturbations without a declared semantic alternative, purely textual hypotheticals, and methods in which visual data are only incidental. This review is guided by four research questions:

RQ1: How should visual counterfactuals be operationally defined and distinguished from related concepts such as adversarial changes and augmentation?

RQ2: How do visual counterfactual methods differ in their technical roles, representations, construction mechanisms, and information or access requirements?

RQ3: How are visual counterfactuals evaluated, and what claims can the reported evidence support regarding their intended roles?

RQ4: What recurring trade-offs, dependencies, and evidence gaps emerge across methodological families and application domains, and what research priorities follow?

By answering these questions, this review makes four contributions: (i) it establishes an operational scope for visual counterfactuals and clarifies the distinction between artifacts and the technical roles of counterfactuals, helping to prevent the uncritical transfer of evidence across different applications. (ii) It introduces a well-structured six-axis taxonomy that differentiates by technical purpose, intervention level, construction mechanism, information and access dependencies, output scope, and target semantics. (iii) It connects evaluation methods to claim–metric relationships, clarifying what common metric families can and cannot achieve. (iv) It assembles a carefully curated, non-redundant collection of 95 primary studies from multiple sources, emphasizing recurring methodological trade-offs, dependencies, and areas for future research.

The remainder of the paper is organized as follows. Section 2 describes the review design, corpus construction, and eligibility criteria. Section 3 establishes the operational definition and separates counterfactual artifacts from their technical roles. Section 4 presents the multidimensional taxonomy. Sections 5–7 present the literature across methodological evolution, technical roles, and application domains, respectively. Section 8 examines evaluation metrics and their evidential limits. Section 9 integrates the principal findings and develops the open research agenda, and Section 10 concludes the review.

2 Review Methodology

2.1 Review Design and Protocol

We conducted a systematic scoping review with technical analysis to examine visual counterfactual methods across explanation, learning, auditing, benchmarking, mitigation, and repair. This supports broad coverage and detailed comparison of their assumptions, mechanisms, and evaluation practices. We used explicit eligibility criteria, structured screening, and extraction; each study was a unit of analysis, with related preprints grouped with their published versions.

The search combined automated retrieval via the OpenAlex API and manual searches in Web of Science, ScienceDirect, and IEEE Xplore; the two-stage strategy is summarized in Table 1. First, eight predefined search strings (S1–S8)

Table 1. Search strategy: primary query strings and modular components used for supplementary manual searches.

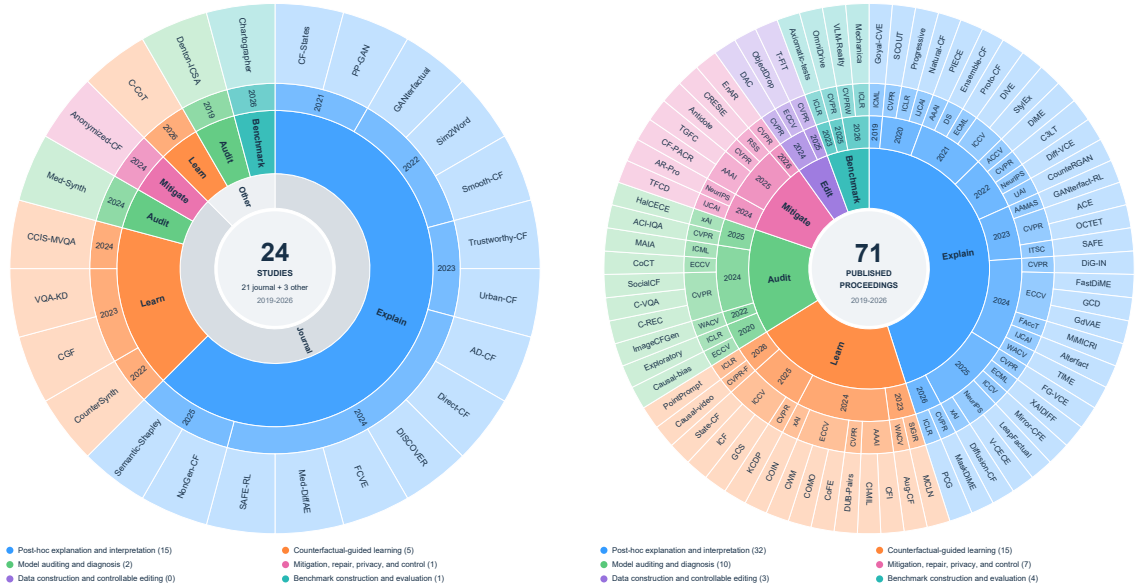
ID	Search string / Component
Primary Search Strings: Predefined Queries	
S1	counterfactual explanation computer vision image
S2	counterfactual visual explanation image classification
S3	counterfactual explanations medical imaging
S4	generative counterfactual explanations computer vision
S5	counterfactual image generation explainable AI
S6	counterfactual explanation object detection segmentation
S7	visual counterfactual explanation deep learning
S8	counterfactual image editing classification explanation
Supplementary Manual Search Components: Boolean Query Blocks	
Core Blocks (combined with every role block below)	
Counterfactual block	counterfactual OR counterfactuals
Visual block	image OR visual OR vision OR video OR multimodal OR "vision language" OR "vision-language" OR VQA OR VLM OR LVLM
Role Block	
Learning	learning OR training OR augmentation OR representation OR debiasing OR "domain generalization"
Question answering	"visual question answering" OR VQA OR "visual reasoning"
Evaluation	benchmark OR dataset OR evaluation OR compositionality OR grounding
Mitigation and repair	hallucination OR mitigation OR correction OR repair OR calibration
Auditing	audit OR auditing OR fairness OR bias OR robustness OR privacy
Generation and causal modeling	generation OR editing OR intervention OR causal OR "structural causal model"

targeted counterfactual explanations across computer vision tasks, including image classification, medical imaging, object detection, segmentation, and image generation. Second, supplementary manual searches extended the coverage to broader applications of counterfactual methods in visual and multimodal settings. These searches combined a counterfactual terminology block and a visual terminology block with one of seven role-specific blocks, covering learning, question answering, evaluation, mitigation and repair, auditing, and generation and causal modeling. Terms within each block were connected using Boolean OR, while the three blocks were combined using Boolean AND. These were supplemented by targeted searches of literature from top AI, computer vision, and machine learning conferences. The search covers publications from 2017 to 2026 (conducted between June 1 and August 31, 2026).

Records retrieved from all sources were pooled and deduplicated using DOI, followed by title, author, and publication-year matching; ambiguous matches were checked manually. The resulting 273 unique records entered eligibility screening (eligibility criteria are discussed in 2.2). Ninety-five (95) studies were retained in the final corpus: 71 conference proceedings, 21 journal articles, and 3 preprints. The composition of the final corpus by conference proceedings, journal articles, and preprints is shown in Figure 1.

2.2 Inclusion and Exclusion Criteria

Studies were eligible when they: **E1:** reported an original method, experiment, dataset, or benchmark; **E2:** used visual or visually grounded data as a central methodological component; **E3:** implemented a counterfactual intervention, construction, comparison, or paired evaluation; **E4:** established an identifiable relationship between a factual case and an alternative outcome, action, state, or representation; and **E5:** made the counterfactual operation central to the contribution.



(a) Journal publications ($n = 21$) and Preprint ($n = 3$).

(b) Conference publications ($n = 71$).

Fig. 1. Distribution of the 95 included studies by publication type and technical role.

Criterion E2 was operationalized using three screening categories: *Visual-primary* checks if studies developed methods specifically for images, video, visual states, or visually grounded data. *Visually substantiated cross-domain* checks if studies proposed general or cross-modal methods but included a substantive visual implementation, evaluation, and visual-specific claim. Both categories were eligible. *Incidental-visual* studies used an image dataset only as an illustrative benchmark for a predominantly non-counterfactual method and were excluded.

Automated screening was designed to apply these rules and was first piloted on twenty deliberately varied studies, including visual-explanation methods, counterfactual-guided applications, and ambiguous boundary cases. The pilot separated technical purpose from representation artifact type and introduced graded visual centrality to distinguish substantive visual counterfactuals from incidental use of image datasets. After this, the screening was applied to the entire corpus.

Final eligibility was verified manually from the study record for retained or ambiguous candidates[1]. Studies were excluded if they failed any eligibility criterion or were surveys, reviews, tutorials, perspectives, or general overviews. Records belonging to the same publication family were consolidated, with the peer-reviewed version retained when available; a preprint was included only when no corresponding peer-reviewed version had been identified by the review cut-off.

3 Definition and Scope

Let $x \in \mathcal{X}$ denote a factual input, let $F : \mathcal{X} \to \mathcal{Y}$ denote the target model, and let q denote the requested target or foil. An intervention is parameterized by $\delta \in \Delta(x)$ and implemented by $\iota : \mathcal{X} \times \Delta \to \mathcal{X}$, producing $x^{\text{cf}}_{\delta} := \iota(x; \delta)$.

[1]The code for automated search and eligibility screening is available here https://github.com/Yinkpeace/visual_counterfactual_literature_search.git

Many optimization-based visual counterfactual methods can be represented by the constrained problem

$$\begin{aligned} \delta^\star \in \arg\min_{\delta \in \Delta(x)} \quad & \lambda_{\text{chg}} D\left(x, x_\delta^{\text{cf}}\right) + \lambda_{\text{plaus}} \Phi_{\text{plaus}}\left(x_\delta^{\text{cf}}\right) \\ \text{subject to} \quad & \mathcal{T}_q\left(F(x), F(x_\delta^{\text{cf}})\right) = 1, \\ & \mathcal{S}_q\left(x, x_\delta^{\text{cf}}\right) \geq \tau_{\text{sem}}, \\ & c_k\left(x, x_\delta^{\text{cf}}\right) \leq \epsilon_k, \qquad k = 1, \ldots, K. \end{aligned} \tag{1}$$

Here, D measures intervention magnitude or cost, Φ_{plaus} is a lower-is-better plausibility penalty, and $\mathcal{T}_q$ specifies the required target-model relation. For a probabilistic reference model, one possible penalty is $\Phi_{\text{plaus}}(x') = -\log p_{\text{ref}}(x')$. The semantic score $\mathcal{S}_q$ encodes a construction-time criterion for the intended semantic change and may be implemented using a surrogate evaluator, while $c_k \leq \epsilon_k$ encodes preservation, feasibility, anatomical, physical, or other domain constraints. When target achievement is optimized softly, the first constraint can be replaced by

$$\lambda_{\text{tar}} \ell_{\text{tar}}\left(F(x), F(x_\delta^{\text{cf}}); q\right)$$

in Equation (1), where $\ell_{\text{tar}} \geq 0$ and vanishes when the target relation is satisfied. Likewise, a hard constraint $c_k \leq \epsilon_k$ can be relaxed using $\mu_k [c_k - \epsilon_k]_+$. An empty feasible set should result in failure or abstention rather than selection of the least invalid candidate.

This formulation highlights the core scientific question in optimization-based methodologies: what defines an intervention, which prior makes an alternative plausible, which content must remain unaltered, and what evidence distinguishes an explanation of the target model from an output-satisfying modification. Section 5 traces how methodological families address these requirements, from explicit foils and semantic representations to generative, domain-constrained, and multimodal interventions.

3.1 Operational Definition and Scope Boundaries

Visual counterfactuals are commonly motivated by the contrastive question "Why outcome y rather than alternative $y^\star$?" [Miller 2021]. We define an *operational visual counterfactual* as an implemented comparison between a factual case and an alternative constructed by changing or controlling one or more visual or visually grounded variables to explain, test, train, or alter a model-relevant outcome. The comparison should identify a factual anchor, a target or foil, the object of intervention, and the resulting counterfactual artifact or controlled pair. This artifact might be an image, region, visual concept, scene, video, or multimodal representation grounded in visuals. Hence, qualification relies on an explicit factual–counterfactual relationship rather than on photorealism, output format, or generator architecture.

Counterfactuals carry distinct implications in explainable artificial intelligence (xAI) and causal inference. In the model-relative formulation associated with [Wachter et al. 2017], a counterfactual explanation identifies an alternative input that produces a desired response from a fixed predictor, typically subject to proximity or other constraints. This establishes how the predictor responds to the alternative input; however, it does not, on its own, establish how an intervention would alter the underlying object or environment. In Pearl's structural causal framework [Pearl et al. 2003], a counterfactual concerns what would have occurred for the same factual case under a specified intervention. Its assessment uses factual evidence to infer background conditions, modifies relevant structural equations, and predicts consequences under those same conditions. This *abduction–action–prediction* process distinguishes case-specific

Table 2. Counterfactual artifacts, their possible technical purposes, and the evidence required to interpret them.

Artifact family	Examples of the returned object	Purposes	Goal-dependent validation emphasis
Contrastive map or localization	Counter-class heatmap, discriminant region, relevance difference	Explanation, interpretation, audit	Localization faithfulness, intervention tests, human model understanding
Edited or generated image	Region substitution, latent edit, GAN/diffusion/flow output	Explanation, audit, augmentation, mitigation	Target response, preservation, semantic correctness, domain validity, user or downstream outcome
Set or trajectory	Diverse alternatives, Pareto set, probability path	Explanation, diagnosis, learning	Coverage, non-redundancy, ordering, stability, selection burden
Latent, concept, or multimodal representation	Prototype direction, causal factor, interacted/uninteracted feature contrast	Interpretation, representation learning, audit	Representational faithfulness, identifiability, ablation, downstream utility
Visual state or structured scene	Atari state, object-level driving scene, action–consequence alternative	Policy explanation, safety analysis, planning	Reachability, dynamics, policy response, scenario validity, safety relevance
Mask, residual, or repaired instance	Counterfactual residual, lesion mask, anomaly repair	Weak supervision, localization, repair	Segmentation or repair performance, preservation, shortcut transfer, domain coherence
Controlled factual–counterfactual pair	Edited chart, minimal image variant, video question pair	Benchmarking, diagnosis, training	Edit correctness, invariant preservation, answer recomputation, format and shortcut controls

counterfactual reasoning from an unconditional intervention query. This review employs visual counterfactuals as an overarching operational concept that encompasses both model-relative alternatives and explicitly causal constructions.

At this point, it is important to differentiate counterfactuals from related concepts, such as *targeted adversarial perturbations*, because both may be constructed through closely related optimization procedures. A specified foil and a changed prediction do not, by themselves, distinguish a counterfactual explanation from a targeted adversarial example. [Browne and Swift 2020] emphasizes the semantic interpretability of the perturbed representation, while [Freiesleben 2022] distinguishes the concepts through their relationship to ground-truth labels and their proximity requirements. We therefore distinguish the construction mechanism from its intended role and the evidence supporting that role. Adversarial optimization may contribute to explanation or auditing, but a prediction-changing perturbation does not automatically yield an interpretable account of the model's behavior. For this review, attack-only studies that report perturbation success without a substantive explanatory or other eligible counterfactual role fall outside the scope; adversarially constructed methods are not excluded solely because of their optimization mechanism.

However, some methods explicitly exploit the overlap between counterfactual and adversarial generation. DVCE [Augustin et al. 2022] uses an auxiliary adversarially robust classifier for cone-regularized diffusion guidance, discouraging trivial non-semantic changes while retaining alignment with the target classifier. ACE [Jeanneret et al. 2023] instead optimizes adversarial perturbations through a diffusion-based filtering process to discourage high-frequency and out-of-distribution changes, followed by inpainting-based refinement to preserve unrelated content. These mechanisms seek to make prediction-changing perturbations more interpretable, but do not independently guarantee semantic correctness or explanatory faithfulness.

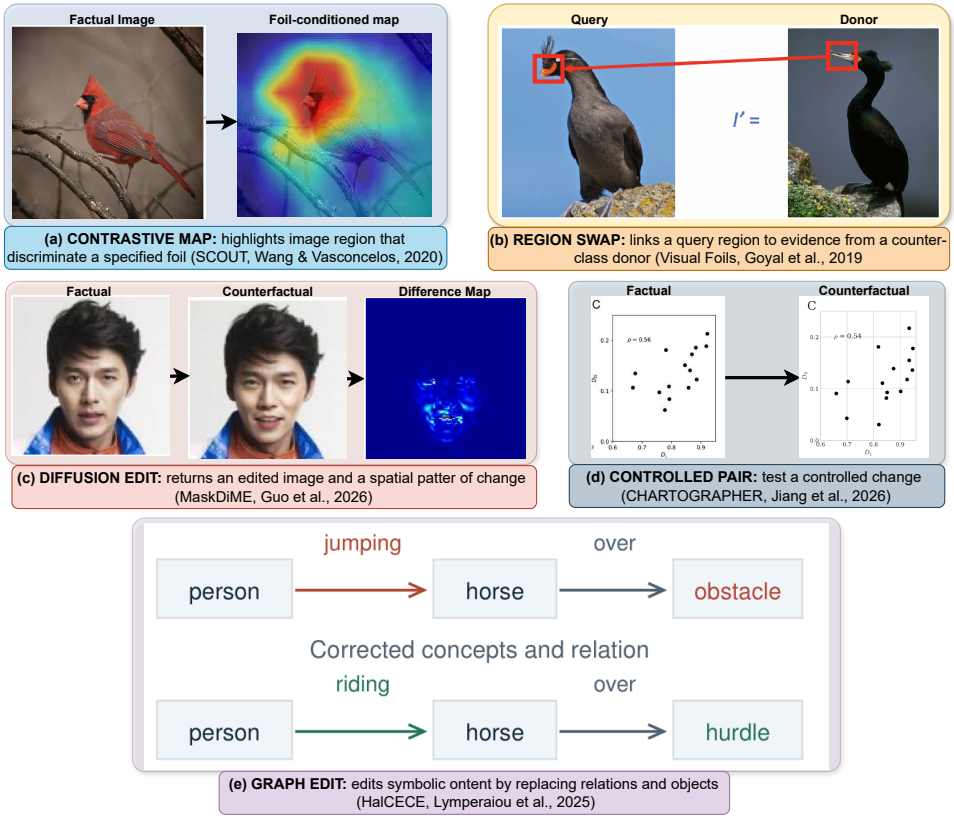


Fig. 2. Representative examples of visual counterfactual artifacts.

3.2 Role–Artifact Separation

The artifact is the output returned by a method, while a role is the purpose of the counterfactual algorithm/mechanism (see Figure 2). The artifact and the role for which a CE is used are independent analytical dimensions. Conflating them creates two recurring errors. The first is *artifact essentialism*: assuming that every generated image is an explanation and that every non-image artifact is not. The second is *role transfer*: treating evidence for one use as validation of another. For example, a generated image that improves calibration may be useful as training data, but provides a poor explanation.

Table 2 illustrates the role–artifact separation. The rows are artifact families, not mutually exclusive method classes, and the roles are possible uses rather than automatic interpretations. The same artifact can move between roles, but each transition creates a new validation obligation.

Several studies illustrate this separation. Counterfactual inpainting can convert an abnormal-to-normal image difference into a weak segmentation mask [Shvetsov et al. 2024]; paired cross-modal counterfactuals can become representation-learning inputs [Singh et al. 2024]; and social-attribute variants can support both bias auditing and mitigation [Howard et al. 2024]. Success at segmentation, learning, or mitigation must not, however, be reported as evidence that the artifact provides a faithful or useful explanation of an individual prediction.

Role–artifact separation supplies the organizing principle for the remainder of the review. Section 4 classifies technical purpose independently of intervention representation and mechanism. Sections 5 and 6 then examine how the same construction can migrate across roles, while Section 8 assigns evaluation requirements to the claim being made. This separation allows the corpus to include emerging counterfactual-guided research without weakening the evidential standard applied to visual explanations.

4 Multidimensional Taxonomy

The taxonomy was created using a hybrid deductive and inductive approach. The initial dimensions stemmed from the review questions and key distinctions in counterfactual research, while their categories were refined through corpus coding to include recurring visual, structured, and multimodal formats. Since a method can serve multiple purposes or combine various mechanisms, the taxonomy coding was multi-label. For summarizing at the corpus level, the main technical goal was based on the study's central thesis and primary claimed endpoint, with secondary substantive roles documented separately. As a result, the axes are analytically distinct but interconnected, rather than orthogonal (see Figure 3).

Technical purpose: This axis identifies the reasons for constructing counterfactuals: explanation, learning, auditing, mitigation, artifact construction, or benchmarking. Purpose determines the required evidence; an effective training example is not necessarily a faithful explanation.

Intervention level: This axis identifies what is manipulated, from pixels and regions to latent representations, concepts, and structured relations. It differs from the resulting artifact: a latent intervention may produce an image. Each level entails distinct preservation and consistency requirements.

Construction mechanism: This axis describes how alternatives are obtained through retrieval, search, learned generation, causal intervention, or programmatic construction. Hybrid methods are distinguished by the component that constructs or selects the alternative and its auxiliary dependencies; generative realism alone does not establish explanatory faithfulness.

Information and access dependencies: This axis records required target-model access, external models, and data during training and inference. Labels, scores, gradients, representations, generators, and semantic oracles impose distinct portability, computational, and reproducibility constraints that black-box or white-box labels alone cannot capture.

Output scope: This axis distinguishes individual alternatives, diverse or Pareto sets, trajectories, and corpus-level patterns or datasets. These outputs require different evidence concerning local validity, coverage, sequential coherence, or generalization.

Target semantics and foil selection: This axis specifies the intended change and how its reference alternative is chosen. Categorical, continuous, structured, and behavioral targets entail different success criteria; foils may be user-specified or automatically selected. Visual counterfactual validity, therefore, extends beyond label flipping.

This multidimensional view supplies the vocabulary for the remaining synthesis. Section 5 uses artifact and mechanism to trace methodological evolution. Section 6 uses technical purpose to examine roles beyond post-hoc explanation. Section 7 adds application-specific invariants, and Section 8 expands the evaluation levels into a claim-sensitive evidence framework.

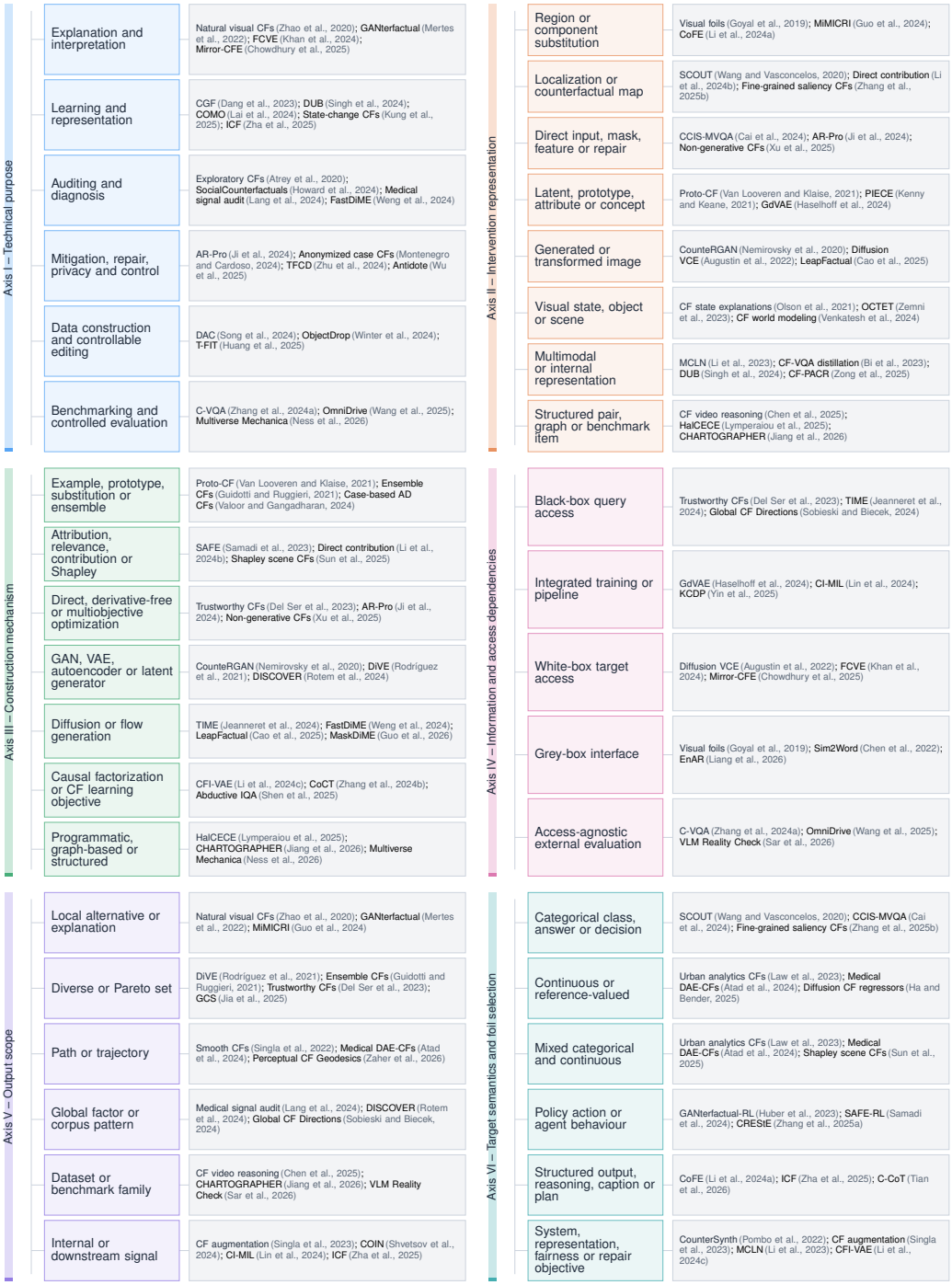


Fig. 3. Multidimensional taxonomy of visual counterfactual methods. The six axes describe complementary properties of each study, and terminal examples are illustrative rather than exhaustive.

5 Methodological Evolution of Visual Counterfactuals

5.1 From Visual Foils to Explicit Contrast

Visual counterfactual research focuses more on defining or uncovering contrasts than on image creation. While an attribution map for class y can show what supports that class, it does not necessarily explain why the model chose y over a specific alternative $y^\star$. Goyal et al. made the contrast explicit by pairing a query image with a distractor predicted as the counter-class and searching for areas where changing them could steer the query towards that class [Goyal et al. 2019]. The region-swap method is contrastive but computationally intensive and tied to candidate substitutions. SCOUT [Wang and Vasconcelos 2020] seeks a discriminant explanation directly from the network that locates features supporting the prediction, features whose absence blocks the foil, and the confidence structure connecting them. Its principal contribution is an optimization-free, class-contrastive localization map that can identify semantically meaningful parts and attributes. SCOUT shows where a contrast resides, but does not construct an attainable counterfactual image.

Substitution and localization represent a fundamental trade-off among pixel- and region-based methods. Substitution requires direct intervention; however, it is limited by donor support and image composition. Localization offers a more efficient and concise visual approach, but it does not achieve the same level of intervention. Considering the feature map as a plausible counterfactual would blur the evidential distinction. Subsequent relevance-based methods face this challenge as well, even when more closely aligned with network internals. Direct-contribution inference [Li et al. 2024a] searches for input components whose alteration significantly impacts the model's output; FCVE [Khan et al. 2024] intervenes on internal filters or concepts and decodes the effects. Mirror-CFE creates trajectories within the classifier's feature space, while fine-grained saliency partition targets object- and part-level evidence associated with misclassification [Chowdhury et al. 2025; Zhang et al. 2025b]. These approaches strengthen representational faithfulness relative to unconstrained image editing. However, their explanations still depend on the contribution estimator, the layer or feature space used, the decoder, and the semantics of the intervention.

Atrey et al. emphasize the importance of this conditionality matters [Atrey et al. 2020]. Rather than solely accepting saliency maps as explanations of the deep reinforcement learning behavior, the researchers interpret the semantic narratives derived from these maps as hypotheses. These hypotheses are then verified through counterfactual modifications to Atari states. Such counterfactual analysis can reveal that a highlighted region that appears intuitive may not genuinely support the asserted environmental significance. Their study demonstrates that visual coherence and narrative plausibility alone are insufficient unless the proposed relationship is substantiated through an intervention test.

This set of work, therefore, establishes *contrast before generation.* A visual counterfactual must identify an alternative, not merely salient evidence for the observed output. Yet an explicit foil does not determine the correct perturbation vocabulary, and a contrastive map does not realize a counterfactual. These limitations motivated semantic representations in which the perturbation could be described as an attribute, prototype, concept, or structured factor rather than an arbitrary region.

5.2 From Contrastive Localization to Semantic Perturbation

Semantic perturbation addresses a failure of pixel and region methods: numerical or spatial minimality does not guarantee a meaningful counterfactual. The central idea is to search in a representation whose coordinates already carry some interpretation. This can make an alternative easier to name and constrain.

Zhao et al. connect the visual foil to an attribute vocabulary [Zhao et al. 2021]. The method modifies the textual attribute description of one class by incorporating attributes associated with another class, generates images from the modified text, and then queries the target classifier to identify influential differences. The counterfactual is expressed as a human-readable attribute substitution and then realized visually. Human-assisted classification provides evidence that some generated attribute contrasts communicate class information. However, the method requires a predefined attribute vocabulary, text-image training data, and a generator capable of isolating each requested attribute. Related editing systems use counterfactual abduction to preserve source-specific content during general or compositional fashion editing [Huang et al. 2025; Song et al. 2024]; here, editing fidelity rather than explanation faithfulness is the primary endpoint.

Prototype-guided methods replace an explicit ontology with representative target examples. Van Looveren and Klaise guide counterfactual search toward class prototypes obtained from encoded examples or class-specific trees [Van Looveren and Klaise 2021]. The prototype supplies a destination and accelerates search relative to unconstrained movement toward any decision-boundary point. However, the prototype is a statistical representative, not an individualized attainable state. The method can therefore produce a counterfactual that is visually coherent but semantically entangled with other class factors.

PIECE provides an additional target-relative semantic rule: to identify features that are *exceptional* under a specified target, and subsequently replace them with counter-class-normal values [Kenny and Keane 2021]. This supports both counterfactuals, for which the output changes, and semi-factuals, for which substantial change remains insufficient to cross the boundary. However, the *exceptional* is defined by a learned latent distribution, which is bounded by the decoder quality. Thus, exceptional-feature replacement can inherit entangled factors and decoder bias even when the output appears coherent. DiVE [Rodríguez et al. 2021] later improves diversity and factor separation. Its disentangled latent search encourages multiple, separated alternatives and penalizes explanations that merely exaggerate or remove the classified attribute. However, similar to [Van Looveren and Klaise 2021] and [Kenny and Keane 2021], its performance depends on a disentangled generator. In addition, the pairwise distance formulation makes semantic independence questionable, e.g., changing age, gender presentation, and hair can remain entangled.

A single counterfactual generator, such as PIECE [Kenny and Keane 2021] or DiVE [Rodríguez et al. 2021], often reflects one distance metric, initialization, and search bias. An ensemble of weak counterfactual explainers trained on different feature and instance samples was introduced in [Guidotti and Ruggieri 2021]. This broadens coverage across different data modalities and explanation objectives, but an ensemble cannot create validity that is not present in its individual components. Autoencoder distortion, implausible base solutions, and incompatible distance assumptions can all survive aggregation, while the larger candidate set increases computational and user-selection burden. Del Ser et al. later make trade-offs explicit through Pareto optimization [Del Ser et al. 2024]. The evolutionary multiobjective procedure balances plausibility, change intensity, and adversarial effectiveness, and returns Pareto Fronts with visibly different trade-offs between fidelity to the original and the strength of the target evidence. This result demonstrates that no single weighted objective fully represents the counterfactual problem (Equation 1). However, this shifts the problem from optimization to selection, as neither the model nor the evaluation identifies the most relevant Pareto-optimal explanation for a given user.

Semantic editing also expands the kinds of model responses that can be explained. Sim2Word [Chen et al. 2023] addresses pairwise visual similarity instead of a single class decision, linking counterfactual similarity regions to representative attribute words. It produces both saliency maps and attribute words that indicate which visible characteristics make two identities appear similar or dissimilar. Law et al. extend counterfactuals to holistic image classifiers and

regressors in urban analytics, where relevant properties are distributed across a streetscape rather than localized to one object [Law et al. 2023]. However, a method can clarify vegetation or building effects yet perform poorly for another visual factor, undermining any general claim that generated images are uniformly more intuitive than heatmaps.

The same semantic machinery can shift from local explanation to global hypothesis discovery.StylEx learns classifier-specific StyleSpace attributes for visual explanation [Lang et al. 2021]; subsequent medical-image analysis uses this approach for interdisciplinary hypothesis discovery [Lang et al. 2024]. DISCOVER learns classification-oriented disentangled factors, generates exaggerated counterfactuals, and uses known features and domain analysis to interpret image-based in-vitro-fertilization models [Rotem et al. 2024]. Global Counterfactual Directions [Sobieski and Biecek 2024] instead identify dataset-level latent directions that alter classifier decisions, extending local generation toward black-box global analysis.

Semantic representations, therefore, address the lack of an interpretable edit vocabulary at the cost of *semantic-prior dependence*. Attributes necessitate ontologies; prototypes require representative geometry; exceptional features demand target-class density models; and disentangled factors depend on identifiability. The unresolved questions concern who defines the semantic variables, the extent of their coverage within the domain, and whether the target model uses them for factual decision-making.

5.3 Generative Manifold Approaches

Once a semantic edit is defined, it still needs to be slightly perturbed. Approaches like GANs, VAEs, autoencoders, and image-translation models have become popular because they restrict modifications to a learned data manifold and optimize within or through the model. This strategy addresses visually unlikely pixel changes and enables amortized generation.

CounteRGAN [Nemirovsky et al. 2020] amortizes target-directed residual generation, whereas C3LT [Khorram and Fuxin 2022] learns cycle-consistent transformations within a pretrained generator's latent space. Both reduce per-instance search while inheriting the generator's representational limits. GdVAE [Haselhoff et al. 2024] integrates classification and closed-form latent counterfactual construction into a self-explainable generative model, trading post-hoc flexibility for assumptions imposed by its class-structured latent space.

The case-based medical explanation study in [Montenegro et al. 2021] demonstrates that counterfactuals can be informative; however, they may also reveal sensitive identity information. Montenegro et al. jointly address realism, privacy, and explanatory value within a generative case-explanation system and incorporate a counterfactual module into the retrieval of factual cases. Nevertheless, empirical de-identification and visual clarity do not constitute formal privacy guarantees, and the removal of identity may also eliminate important diagnostic or explanatory details. Later disentanglement work continues this direction [Montenegro and Cardoso 2024]. Privacy-aware counterfactuals expose a general problem: invariants can conflict, and preserving explanation while suppressing identity is not reducible to image quality alone. GANterfactual [Mertes et al. 2022] uses cycle-consistent abnormal-to-normal translation to give medical non-experts full-image contrasts and compares them with saliency-based explanations. The resulting human evidence (from non-experts) supports improved satisfaction outcomes for the tested population. However, non-experts cannot determine whether anatomy is preserved, and cycle consistency does not enforce that only pathology changes.

Progressive Exaggeration generates image sequences spanning classifier confidence levels [Singla et al. 2019]; related chest-radiograph counterfactuals use probability-conditioned paths to convey evolving diagnostic evidence [Singla et al. 2023a]. The same trajectories can be repurposed as training data, as shown in [Singla et al. 2023b]. Augmentation through Counterfactual Explanation involves assigning soft labels along counterfactual pathways and refining an

overconfident classifier. This process subsequently enhances calibration accuracy, out-of-distribution performance, and overall robustness. However, the utilization of a shared generator and datasets inherently introduces correlated failure modes in both explanation and augmentation procedures. Furthermore, factors such as wiring, acquisition artifacts, and correlated anatomical structures may influence the diagnosis.

Generative methods are also used to explain reinforcement learning (RL) models. Olson et al. generate minimally changed Atari states that cause an RL agent to choose a different action [Olson et al. 2021]. Their user studies test whether non-experts can distinguish between generated and real game frames and identify a flawed agent. The state generator, however, models visual similarity rather than guaranteed environment reachability. The GANterfactual-RL [Huber et al. 2023] reframes visual state generation as a domain-translation process over action outcomes, assessing whether alternative options help users differentiate agent strategies. Nevertheless, this approach also incorporates translation assumptions, suggesting that action-conditioned domains may contain multiple states and extended contextual information, thereby indicating that translating the current frame could conflate immediate evidence of actions with strategic behavioral indicators.

Across these studies, the generator plays at least four roles: (i) a density prior, (ii) a renderer, (iii) an amortized optimizer, and (iv) an edit vocabulary. These roles are often evaluated jointly, although no single metric can verify that a generator preserves the factual instance's latent causes (Section 8). Mode coverage is as critical as visual fidelity because if the generator cannot express a valid alternative, optimization will return the nearest available surrogate, potentially altering an unintended correlated factor. Consequently, the generative-manifold stage enhances *visual realizability* while establishing *manifold authority*. The learned model determines which counterfactual scenarios are accessible. While improved architectures can reduce artifacts and improve coverage, they are insufficient on their own to determine which variables should be mutable, identify causal dependencies, or select the appropriate alternative responses to the user's foil.

5.4 Diffusion, Flow, and Modern Generative Priors

Diffusion models changed the quality and coverage of image priors available to counterfactual methods. Their iterative denoising process permits classifier guidance, preservation regularization, and spatial control at a scale difficult for earlier task-specific GANs.

Early diffusion-based VCEs include classifier-guided DVCE [Augustin et al. 2022], target-gradient-guided DiME [Jeanneret et al. 2022], and ACE, which regularizes adversarial search through diffusion [Jeanneret et al. 2023]. These methods improve visual plausibility while retaining dependencies on classifier guidance and generative priors. DiG-IN [Augustin et al. 2024] broadens guided diffusion from individual counterfactuals to classifier disagreement, systematic errors, and neuron analysis, whereas FastDiME [Weng et al. 2024] uses self-optimized masking to add or remove suspected shortcuts with a lower inference burden. TIME [Jeanneret et al. 2024] and MaskDiME [Guo et al. 2026] address access and localization, respectively. TIME distills class or context information for a text-to-image model and queries the target as a black box, reducing gradient dependence and latency. A large pretrained diffusion model provides a broad image prior, but the distilled class embedding may reproduce broad correlations rather than evidence for a specific prediction. MaskDiME, on the other hand, introduces adaptive dual masks recomputed during reverse diffusion. Complementary gradient-derived masks control source preservation and target guidance, yielding $O(T)$ complexity rather than nested optimization over diffusion steps. This efficiency retains a white-box requirement and remains sensitive to noisy classifier gradients.

Medical diffusion-autoencoder methods adopt a different compromise. Atad et al. explore semantic codes to generate alternatives for binary classification, ordinal severity, and regression [Atad et al. 2024]. The autoencoder provides a data-specific representation that captures gradual target variation, thereby broadening the range of output types addressed by visual counterfactuals. However, the experiments are restricted to 2D images, and linear or ordinal traversal cannot represent non-monotonic disease progression. Adjacent anatomy can also change with the target, so a smooth trajectory is not a patient-specific causal forecast. Ha and Bender [Ha and Bender 2025] explicitly clarify regressor targeting and compare pixel-space with latent diffusion. Latent diffusion enhances control over targets and the distributional quality, while pixel diffusion may produce sparser results or better maintain certain identity cues. However, the evidence is limited to a synthetic task and perceived age, with the oracle being a model proxy; as a result, factors like age, glasses, expression, identity, and gender presentation might still be confounded. Neither study demonstrates causal monotonicity.

A valid visual counterfactual may cross a classifier's boundary while remaining on the original side of a semantic or human boundary. This failure is particularly damaging when counterfactuals are used for augmentation. LeapFactual [Cao et al. 2026] uses conditional flow matching to move toward a target manifold while emphasizing semantic reliability. Its analysis links counterfactual quality to downstream augmentation and shows that semantically unreliable alternatives can harm learning. However, conditional flow matching is computationally demanding in high-dimensional latent spaces, and imbalanced data can distort the target transport.

Conventional visual counterfactuals require a GAN, a VAE, a diffusion decoder, or a white-box generator tied to a data domain. Xu et al. challenge this assumption through class-conditional denoising scores and derivative-free Bayesian search [Xu et al. 2025]. Although this removes the need for a conventional decoder, it retains dependence on learned distributional score models. Perceptual Counterfactual Geodesics takes a different route by replacing flat pixel or latent distances with paths induced by robust visual features [Zaher et al. 2026]; the resulting semantics remain conditional on the selected feature geometry.

Diffusion and flow improve the limited image prior of earlier generators; however, the primary challenge shifts from *can a convincing image be rendered?* to *which guidance, oracle, and invariant make that image an explanation?*

5.5 Structured and Domain-Constrained Interventions

Generic image realism is weakest where the domain imposes relationships that common perceptual metrics do not encode. Medical anatomy, demographic morphology, object interactions, driving dynamics, and anomaly repair require constraints on how variables change together. Structured methods attempt to replace a generic manifold with a more explicit model of allowable intervention.

CounterSynth [Pombo et al. 2023] uses conditional deformations to generate morphology-preserving 3D brain images under demographic conditions. The topology-preserving transformation is a significant advance over unconstrained texture editing, and the counterfactuals are evaluated as augmentation for settings with imbalance, confounding, or inequitable subgroup performance. However, the generator estimates statistical morphological variation and may not separate demographic effects from site, intensity, pathology, or other confounders.

Domain constraints can also be elicited interactively rather than fully encoded, as shown in MiMICRI [Guo et al. 2024], which segments cardiovascular images into named regions and allows region-level swaps to support expert hypothesis testing. The framework was validated by a small pool of experts, who found the tool useful for examining which anatomical regions influence predictions and for comparing subgroups. However, the two experts cannot establish general usability. The case-based diagnostic maps for Alzheimer's disease presented in [Valoor and Gangadharan

2025] similarly aim to make disease-related change visible. Such maps can compare favorably with generic saliency and support intuitive diagnosis narratives. However, donor swaps can violate anatomy, acquisition consistency, and patient-specific physiology, while segmentation errors define the intervention boundaries.

Counterfactual inpainting turns a structured counterfactual intervention into supervision. COIN [Shvetsov et al. 2024] generates an abnormal-to-normal image and interprets the residual as a lesion mask for weakly supervised medical segmentation. A perturbation/inpainting generator edits an abnormal 2D CT slice until a fixed classifier predicts normal. However, the classifier decides what is erased, relying on a mask that predicts a shortcut aligned with the classifier could be misleading in medical contexts. Additionally, slice-by-slice 2D processing overlooks the volumetric continuity, and thresholding or post-processing can significantly affect the IoU.

Object-centered counterfactual scenes address a parallel problem in autonomous driving. OCTET [Zemni et al. 2023] encodes scenes into an object-structured latent space so that position, appearance, and other object-level factors can be manipulated, and it extends beyond classification to semantic segmentation. SAFE [Samadi et al. 2023] and SAFE-RL [Samadi et al. 2024] use saliency to focus counterfactual changes on evidence relevant to automated-driving systems and visual RL policies. ObjectDrop [Winter et al. 2024] learns the broader-scene effects of object insertion and removal from paired counterfactual captures; Counterfactual World Modeling and point-prompted video generation extend controlled visual alternatives to physical dynamics representations and motion tracking [Shrivastava et al. 2026; Venkatesh et al. 2024]. These approaches add object or temporal structure, but their validity remains bounded by the assumptions of decomposition, generation, and dynamics. Semantic Shapley interventions offer another route to scene structure by assigning contributions to meaningful components before generating counterfactual driving scenarios [Sun et al. 2025]. This supports component-level attribution and alternative-scene analysis for end-to-end driving. However, the attribution depends on DeepSHAP's black baseline and approximation; semantic values inherit segmentation error and can be biased by region size.

These studies reveal at least four levels of domain structure: (i) *Representational structure*, such as named anatomy, objects, or semantic components. (ii) *Statistical structure*, such as population morphology or a learned distribution of normal images. (iii) *Relational and dynamic structure*, such as physiological coupling, occlusion, transition reachability, and multi-agent interaction. (iv) *Causal and institutional structure*, such as interventions available to an individual, acquisition processes, privacy obligations, and safety consequences.

Most domain-centered visual methods reach the first two levels and only partially explore the third. Very few reach the fourth. This highlights the importance of accurately describing their validity: anatomy-aware methods are more preferred to generic realism, and object-aware approaches are better than unconstrained scene editing. However, neither approach guarantees a causal, counterfactual understanding of the world.

5.6 Visual and Multimodal Reasoning Interventions

This subsection presents the latest methodological shift, which involves changing the counterfactual objects. For instance, in visual language models (VLMs) or visual Q&A (VQA) systems, the relevant alternatives might be a masked region, a multimodal representation, a caption concept, or action-consequence narratives. These approaches do not have to produce an explanation image; instead, they can use a controlled visual relation for training, testing, or guiding reasoning.

Counterfactual VQA methods employed visual masking and paired supervision to reduce dependence on language priors. Bi et al. combine counterfactual examples with self-supervised teacher-student distillation, preserving relations between factual and counterfactual question–image pairs [Bi et al. 2024]. The counterfactual is valuable as a training

relation, and improvement on shifted VQA benchmarks supports that role. CCIS-MVQA [Cai et al. 2024] integrates layer-wise relevance propagation, masked visual interventions, and causal-effect-inspired training for medical VQA. Other methods construct multimodal hard negatives for concept composition, fine-grained referring-expression mismatches, or counterfactual reasoning chains [Lai et al. 2024; Yu and Li 2024; Zhang et al. 2024a]; their contributions can be measured in terms of task-specific supervision rather than explanation faithfulness.

Counterfactual learning can also operate entirely in multimodal representation space. MCLN [Li et al. 2023] contrasts content from interacted and uninteracted items to identify preference-irrelevant visual and textual features and improve multimedia recommendation. Graph-based collaborative representations are fused with image and text embeddings. However, an uninteracted item may go unseen rather than rejected, so exposure bias can confound the implied preference counterfactual. Thus, learned differences are not necessarily causal effects, and the method does not present users with visual alternatives. The Counterfactual Generation Framework presented in [Dang et al. 2023] similarly separates class-specific and sample-specific information and recombines them to synthesize training examples, thereby improving scarce-data classification on the tested benchmarks.

Video reasoning turns counterfactual structures into both a benchmark and a post-training signal. Chen et al. derive causal-graph-guided counterfactual questions and use supervised and reinforcement-based post-training to improve graph-consistent video answers [Chen et al. 2026]. HalCECE [Lymperaiou et al. 2025] represents image-caption hallucination through conceptual counterfactuals on scene-like structures. VLM Reality Check [Sar et al. 2026] shows that aggregating multimodal accuracy can hide sensitivity to visual framing, demographic cues, language, and presentation. However, synthetic transformations may not preserve semantics equally across cultures, languages, or image classes. Therefore, controlled sensitivity is not automatically a causal cognitive bias unless the intervention and invariant label are justified by the data-generating process.

Counterfactual chain-of-thought (C-CoT) [Tian et al. 2026] makes alternative actions and consequences explicit within autonomous-driving visual reasoning. Visual observations are converted into structured semantic relations, followed by critical-object identification, risk prediction, counterfactual risk reasoning, and action planning. However, counterfactual labels are derived from post-intervention rules, so benchmark performance partly measures agreement with an engineered annotation process. Finally, EnAR [Liang et al. 2026] uses counterfactual visual impression to address VLM's hallucination. It uses a diffusion prior to construct a normalized counterfactual *visual impression*, contrasts attention between factual and normalized images, identifies tokens associated with the discrepancy, and prevents those tokens from driving generation. However, the diffusion prior may erase a genuinely rare event, thereby making the model confidently conventional rather than correct.

Chart VLMs might produce responses based on familiar layouts, outdated priors, or OCR shortcuts instead of the actual plotted data. CHARTOGRAPHER [Jiang et al. 2026] reverse-engineers a chart into executable code, verifies the reconstruction, modifies underlying values, and creates multiple controlled visual counterfactuals. A model translates chart images into code and data; automated checks and human validation filter out poor reconstructions. Using different random seeds, it produces variants in which data values change while the overall chart style remains consistent.

These multimodal methods share a methodological principle: construct paired worlds that differ in a controlled visual or visually grounded variable, then use the response difference as supervision, diagnosis, evaluation, or control. Their main challenge is *relational validity*, that is, how to determine if the pair maintains all relevant facts, alters the intended fact, produces the correct consequence, and is resistant to shortcuts caused by the construction process.

This section outlines a progressive methodological trend: starting with an implicit counterfactual contrast and moving toward an explicit foil; transitioning from arbitrary modifications to explicitly named factors; evolving from

unrendered alternatives to learned worlds; shifting from generic image priors to structured constraints; and progressing from explanation artifacts to counterfactual relations integrated throughout a system. The outstanding requirement is that a counterfactual change should alter the target for the intended reason, preserve the factual case under relevant semantics, remain credible within the domain structure, and enhance the desired human or system outcome.

6 Various Technical Roles Assumed by Visual Counterfactuals

The expansion of visual counterfactuals beyond post-hoc explanation has expanded the roles and changed the beneficiary, the intervention point, the outcome variable, and the evidence required for success.

Let $a = C(x, F, q)$ denote a counterfactual artifact constructed for factual case x, target system F, and contrast q. In this formulation, the quality of the counterfactual artifacts varies depending on their specific role, since an artifact might be highly valuable in one context but less so in another. Accordingly, this section examines the utility evidence relevant to each role.

6.1 Counterfactual-Guided Learning

Counterfactual-guided learning uses a factual–counterfactual relation to change the training distribution, representation, supervision, or objective. It differs from ordinary augmentation in that it adds examples relative to a model outcome, a target factor, or a controlled intervention. It also differs from post-hoc explanation because the primary object of evaluation is what the model learned after exposure to the counterfactuals.

Boundary-focused augmentation. Augmentation by Counterfactual Explanation [Singla et al. 2023b] reuses probability-conditioned paths around a classifier boundary as soft-labeled training data. The method targets accurate but overconfident classifiers whose training data provide insufficient support near ambiguous in-distribution regions and outside the original classes. The counterfactual paths, therefore, sample regions the current model regards as transitional. CounterSynth [Pombo et al. 2023] uses a more structured augmentation mechanism for brain imaging, employing conditional deformations to produce images under alternative demographic conditions to address class imbalance, confounding, and inequitable subgroup performance. The method illustrates a crucial distinction between *counterfactual augmentation* and an *individual causal counterfactual*. The Counterfactual Generation Framework [Dang et al. 2023], on the other hand, separates sample-specific from class-related information and recombines these factors for scarce-data learning. Finally, LeapFactual [Cao et al. 2026] demonstrates that corresponding augmentation risks, such as target-valid but semantically unreliable alternatives, can damage downstream learning.

Representation-level counterfactual learning. MCLN [Li et al. 2023] compares multimodal content from interacted and uninteracted items to suppress components considered irrelevant to user preference. The factual–alternative relation acts as a representation filter. Other approaches use counterfactual intervention to regularize causal visual representations or generate hard negative sets for vision–language composition [Jia et al. 2025; Lai et al. 2024; Li et al. 2024c]. VQA methods [Bi et al. 2024] use counterfactuals to weaken language shortcuts and improve visual grounding, while CCIS-MVQA [Cai et al. 2024] uses relevance-guided masking to estimate answer sensitivity and train a medical VQA model. Across audiovisual tasks, factual–counterfactual contrasts also support representation learning, segmentation, physical-commonsense debiasing, and short-video inference [Singh et al. 2024; Zha et al. 2025; Zong et al. 2025a,b]. Related objectives address multimodal sarcasm and cross-domain visual-emotion recognition [Yin et al. 2025; Zhu et al. 2024].

Video counterfactual reasoning moves the supervision from isolated masks to structured event relations. Chen et al. use causal graphs to generate factual-counterfactual reasoning tasks and apply supervised and reinforcement-based post-training [Chen et al. 2026]. State-change counterfactuals [Kung et al. 2025] instead supervise procedure-aware video encoders with counterfactual failure outcomes. C-CoT integrates counterfactual alternatives into visually grounded planning [Tian et al. 2026]. While these evaluations demonstrate the models' utility within their specific task families, they do not indicate that the models have developed a general intervention approach to physical data, since the graphs, descriptions, question formats, and objectives can all be learned as task-specific patterns.

Counterfactuals as weak supervision. COIN [Shvetsov et al. 2024] turns an abnormal-to-normal counterfactual difference into a segmentation label. Counterfactual inference additionally provides pseudo-labels for instances in whole-slide multiple-instance learning, whereas patch-swapped factual–counterfactual contrasts facilitate radiology report generation [Li et al. 2024b; Lin et al. 2024]. Their downstream performance assesses whether the constructed relation provides useful supervision, rather than whether it explains the original model. If the guiding classifier depends on a shortcut, the generated label or contrast can faithfully supervise incorrect evidence; therefore, shortcut audits and domain validation are necessary alongside task metrics.

In summary, these investigations demonstrate that counterfactuals have the potential to enhance model representations and performance; however, they also reveal errors that occur when evidence is erroneously transferred to incorrect claims. A comprehensive evaluation of learning should address four crucial questions: (i) Are the factual and counterfactual pairs semantically accurate at the sample level? (ii) Do they contribute additional information beyond merely increasing computational resources, parameters, or generic augmentation? (iii) Does the observed improvement remain consistent across different shifts, subgroups, and seeds used for generating counterfactuals? (iv) When these artifacts are also intended as explanations, what independent evidence substantiates this role?

6.2 Auditing and Diagnosis

Counterfactual auditing uses an intervention to test a hypothesis about a model or dataset. Its primary output is evidence that a proposed factor matters, does not matter, or interacts with other factors. Let z denote a specified intervention, let $C_z(x)$ contain the alternatives that implement z while satisfying its required invariants, and let $s_q : \mathcal{Y} \to \mathbb{R}$ select the target response of interest. The local model-response contrast is

$$\Delta_{F,q}(z;x) := s_q\Big(F(x_z^{\mathrm{cf}})\Big) - s_q(F(x)), \qquad x_z^{\mathrm{cf}} := \iota_z(x) \in C_z(x). \tag{2}$$

The functional s_q may represent a class score, an anomaly score, a value estimate, an action probability, or a task-specific structured-output score. Interpreting $\Delta_{F,q}$ requires more than merely observing a changed output or image: ι_z must modify the intended factor, preserve relevant invariant features, and avoid cues that could suggest an alternative explanation for the response. Note that Equation 2 defines a model-response contrast, which is not necessarily indicative of a causal effect; causal interpretation further requires a justified structural intervention and the corresponding identification assumptions [Adedigba et al. 2026; Karimi et al. 2021]. For stochastic construction, the initial term would be averaged over the randomness inherent in generation and accompanied by an appropriate measure of uncertainty.

Falsifying explanatory hypotheses. Atrey et al. begin from saliency-derived claims about Atari agents and apply counterfactual state changes to determine whether those claims survive empirical testing [Atrey et al. 2020]. Attribution-guided methods such as SCOUT, FCVE, direct-contribution inference, SAFE, and SAFE-RL similarly use class-discriminant maps, internal concepts, contribution estimates, or saliency to select candidate evidence [Khan et al.

2024; Li et al. 2024a; Samadi et al. 2024, 2023; Wang and Vasconcelos 2020]. Their interventions can strengthen the case that the highlighted evidence is model-relevant, but only if the guide and perturbation are tested independently. When the same attribution both selects the intervention and supplies the faithfulness metric, the audit can become self-confirming.

Discovering global factors and shortcuts. Lang et al. use generative latent factors and interdisciplinary review to formulate hypotheses about visual signals learned from medical imagery [Lang et al. 2024]. Similarly, DISCOVER [Rotem et al. 2024] learns classification-oriented factors and compares them with known features in in-vitro fertilization imagery. MiMICRI [Guo et al. 2024] illustrates domain-expert auditing as a formative process. Experts interact with segmented cardiovascular regions and identify physiologically implausible independent swaps, while FastDiME [Weng et al. 2024] adds and removes suspected shortcut features to quantify their influence. These workflows broaden auditing from one prediction to a recurring model and dataset structure. They can reveal acquisition artifacts, demographic patterns, or biologically meaningful morphology that saliency maps do not express. The multiobjective methods can also support auditing by exposing triviality, bias, and several ways in which a prediction changes, including concept-related alternatives omitted by the nearest solution [Del Ser et al. 2024; Rodríguez et al. 2021].

Diagnosing hallucination and grounding. HalCECE [Lymperaiou et al. 2025] constructs conceptual counterfactuals to explain whether an image-caption mismatch involves an extra object, a missing object, or an incorrect relation. CHARTOGRAPHER [Jiang et al. 2026] and VLM Reality Check [Sar et al. 2026] extend diagnosis across controlled families rather than individual captions. These benchmark roles are discussed further below, but they also constitute audits because they localize failure to specific changed factors.

6.3 Benchmark Construction and Controlled Evaluation

A counterfactual benchmark transforms the factual-counterfactual relationship into a repeatable test case. For a pair (x_i, x_i^{cf}) with factual answer y_i and recomputed answer y_i^{cf}, define

$$b_i(F) := \mathbf{1}[\mathrm{Correct}_i(F(x_i), y_i)],$$
$$r_i(F) := \mathbf{1}\left[\mathcal{R}_i\left(F(x_i), F(x_i^{\mathrm{cf}}); y_i, y_i^{\mathrm{cf}}\right) = 1\right],$$

where $b_i(F)$ is the model's factual correctness and $\mathcal{R}_i$ encodes the required answer change, preservation, or selective update. Counterfactual consistency, conditional on factual correctness, is then

$$\widehat{C}_{\mathrm{cf|fact}}(F) := \frac{\sum_{i=1}^{n} b_i(F)\, r_i(F)}{\sum_{i=1}^{n} b_i(F)}, \qquad \sum_{i=1}^{n} b_i(F) > 0. \tag{3}$$

For ordinary classification, $\mathrm{Correct}_i$ may denote exact label equality, and $\mathcal{R}_i$ may require $F(x_i^{\mathrm{cf}}) = y_i^{\mathrm{cf}}$. For structured tasks, $\mathcal{R}_i$ should require that the intended component change correctly while specified unaffected components remain unchanged. Conditioning prevents a model that fails both members from appearing consistent for the wrong reason, but factual accuracy and the denominator in Equation 3 should also be reported.

Several construction paradigms appear in the recent literature. Causal-graph video tasks [Chen et al. 2026] specify event relations and generate hypothetical questions whose answers follow from the graph; C-VQA [Zhang et al. 2024b]

constructs direct and indirect hypothetical premises over real and synthetic scenes; OmniDrive [Wang et al. 2025] evaluates candidate ego policy trajectories against structured 3D constraints; and Multiverse Mechanica [Ness et al. 2026] renders aligned worlds from a known game-engine graph. CHARTOGRAPHER [Jiang et al. 2026] reconstructs a chart as executable code, edits the underlying data, rerenders the figure, and recomputes the answer, whereas VLM Reality Check [Sar et al. 2026] applies deterministic minimal visual transformations across multiple bias dimensions and languages. Each design trades breadth against the provenance of its intervention and expected response.

These benchmarks address a limitation of fixed test sets by generating controlled counterfactuals that verify whether the response aligns with the altered visual fact. These counterfactuals can uncover failures obscured by aggregate factual accuracy and enable within-instance comparisons that mitigate some content variability. Counterfactual benchmarks also introduce a training–evaluation boundary issue. Once the benchmark or its generator is used for post-training, performance may improve due to increased familiarity with the construction grammar.

6.4 Mitigation, Repair, Privacy, and Safety

Mitigation methods place the counterfactual within a system to reduce the likelihood of failure. Let S_0 denote the original system; let $m \in \mathcal{M}_{\text{adm}}$ denote an admissible counterfactual-based mitigation strategy, such as an augmentation rule, repair transformation, training intervention, or inference-time controller; and let $S_m := \Gamma(S_0, m)$ denote the resulting system. A general mitigation objective is

$$\begin{aligned} m^\star \in \arg\min_{m \in \mathcal{M}_{\text{adm}}} \quad & \mathcal{R}_{\text{failure}}(S_m; \mathcal{D}_{\text{eval}}) \\ \text{subject to} \quad & \mathcal{U}_k(S_m; \mathcal{D}_k) \geq \mathcal{U}_k(S_0; \mathcal{D}_k) - \epsilon_k, \qquad k = 1, \ldots, K. \end{aligned} \tag{4}$$

Here, $\mathcal{R}_{\text{failure}}$ quantifies the specified failure, such as hallucination, identity leakage, subgroup error, anomalous behavior, or unsafe action. Each $\mathcal{U}_k$ quantifies a property that should be preserved, and ϵ_k specifies its maximum allowable degradation relative to the original system. The admissible set $\mathcal{M}_{\text{adm}}$ encodes feasibility, cost, privacy, or deployment constraints. Evaluation should use held-out data; for level repair, $\mathcal{D}_{\text{eval}}$ may reduce to the factual instance and its validated neighborhood.

Privacy-preserving counterfactuals. Montenegro et al. make privacy an explicit objective of case-based medical explanation while attempting to retain realism and explanatory value [Montenegro et al. 2021]. Their follow-up work uses disentanglement to anonymize explanations of factual and counterfactual cases [Montenegro and Cardoso 2024]. This role requires at least three distinct evaluations: identity leakage, preservation of task-relevant evidence, and explanation utility. Visual de-identification or reduced recognition accuracy under one adversary is not a universal privacy guarantee.

Counterfactual repair. AR-Pro [Ji et al. 2024] treats a counterfactual as a repaired instance satisfying formal properties relative to an anomaly detector. COIN's abnormal-to-normal residual and segmentation use a related preservation problem [Shvetsov et al. 2024]. Normalization can localize pathology, but it can also remove an acquisition artifact or shortcut that the classifier mistook for pathology.

Hallucination mitigation. Antidote [Wu et al. 2025] targets both counterfactual-presupposition and object-perception failures in large vision–language models, while EnAR [Liang et al. 2026] creates a diffusion-based visual counterpart, compares it with the original through the model's visual pathway, and suppresses tokens associated with counterfactual

Table 3. Summary of the Technical Roles of Counterfactual.

Role	Unit benefiting from the counterfactual	Minimum central evidence	Invalid inference to avoid
Learning	Trained model or representation	Downstream gains, ablations, shift/subgroup tests, and sample-level semantic audit	Better model performance implies good explanations
Auditing	Hypothesis about a model or dataset	Valid intervention, stable response effect, and independent semantic corroboration	A changed prediction proves the named causal mechanism
Benchmarking	Evaluation instrument and tested model	Pair correctness, invariant control, recomputed ground truth, leakage tests, and conditional consistency	Benchmark accuracy proves general counterfactual reasoning
Mitigation/repair	Deployed model pipeline or repaired instance	Failure reduction, preservation of valid content, side-effect analysis, and domain feasibility	Removing the measured failure makes the system safe
Privacy	Data subject and explanation recipient	Threat-model-specific privacy, identity suppression, diagnostic preservation, and explanation utility	An anonymized-looking image guarantees privacy
Safety control	Agent, environment, and affected stakeholders	Dynamic validity, scenario coverage, policy consequences, and calibrated residual risk	A plausible edited scene certifies policy safety

conflict. Improvements in hallucination and general tasks support their respective roles in post-training and inference-time mitigation.

Fairness and subgroup mitigation. Fairness auditing predates recent counterfactual augmentation: Denton et al. [2019] probe classifier sensitivity through generated face-attribute changes, and Balakrishnan et al. [2021] uses human-validated synthetic attribute transects. Dash et al. [2022] combine SCM-constrained generation with bias evaluation and counterfactual regularization. CounterSynth and SocialCounterfactuals extend intervention-based analysis and mitigation to brain imaging and intersectional vision–language settings [Howard et al. 2024; Pombo et al. 2023]. Conclusions remain conditional on edit validity, subgroup coverage, and the adopted fairness criterion.

Safety-oriented visual counterfactuals. SAFE [Samadi et al. 2023], SAFE-RL [Samadi et al. 2024], OCTET [Zemni et al. 2023], semantic Shapley scenes [Sun et al. 2025], and C-CoT [Tian et al. 2026] all operate in safety-relevant driving or embodied settings. CREStE instead uses counterfactual trajectory guidance and human queries during long-horizon mapless navigation learning [Zhang et al. 2025a]. These systems respectively explain decisions, edit scenes, identify influential components, reason over alternative consequences, or guide policy learning.

6.5 Role Transitions and Hybrid Methods

The most significant recent advancement is that counterfactuals play a hybrid and interdependent role. Counterfactual artifacts can move between explanation, auditing, learning, and mitigation. Examples include probability-conditioned trajectories reused for calibration [Singla et al. 2023a,b], shortcut insertion/removal [Weng et al. 2024], abnormal-to-normal residuals used for segmentation [Shvetsov et al. 2024], compositional vision-language learning [Jia et al. 2025], and graph-based tasks used for post-training [Chen et al. 2026; Kung et al. 2025]. Each transition requires evidence appropriate to the new role. Studies should identify who or what consumes the artifact, whether its content or label

changes, and which errors may propagate downstream. When construction and evaluation share the same model, independent semantic validation is particularly important.

7 Application-Centred Synthesis

The meaning of a visual counterfactual is partly determined by its application. This section, therefore, synthesizes different application regimes. Across applications, an artifact can satisfy target validity while failing the domain question that gives the target meaning.

7.1 General and Fine-Grained Image Classification

General image classification offers explicit foil classes and established testbeds. Early works were therefore able to isolate foundational questions, such as whether an explanation should identify evidence for the predicted class, evidence for a specified alternative, or an explicit visual difference between them. Goyal et al. use an explicit factual–counterfactual difference to evaluate fine-grained bird classification and machine teaching [Goyal et al. 2019]. SCOUT [Wang and Vasconcelos 2020] localizes evidence that discriminates the predicted class from a counter-class. Attribute-conditioned generation made the foil more communicable by coupling textual class differences with synthesized images [Zhao et al. 2021], while Sim2Word [Chen et al. 2023] connected counterfactual similarity changes to representative attribute words. Together, these studies established that the foil must be explicit: "why this class?" and "why this rather than that class?" need not select the same evidence.

Why classification is a productive comparison regime. Public datasets reveal methodologies relevant to visual architectures of varying complexity. CelebA offers identifiable semantic attributes and an accessible generative domain; CUB emphasizes fine-grained contrast; and ImageNet evaluates scale and diverse class structures. These datasets have facilitated the development of residual, prototype-guided, global-direction, diffusion-guided, classifier-space, and perceptual-geodesic approaches under differing assumptions about access and generation [Augustin et al. 2024; Chowdhury et al. 2025; Nemirovsky et al. 2020; Sobieski and Biecek 2024; Van Looveren and Klaise 2021; Zaher et al. 2026].

Fine-grained contrast and semantic precision. Fine-grained tasks provide a stricter preservation assessment than visually distinct class changes, as most of the image remains unchanged. In bird recognition, explanations should detail differences in morphology, coloration, or markings, without changing pose, background, or unrelated features. Pedagogical research shows such contrasts help users distinguish classes effectively [Goyal et al. 2019; Zhao et al. 2021].

Continuous Image Regression. Image regression changes foil semantics; instead of choosing a different class, the user sets a target value, interval, direction, or amount. Law et al. [Law et al. 2023] propose options for urban image classifiers, showing that some perturbed concepts are interpretable, while others require scene shifts. Diffusion counterfactuals [Atad et al. 2024] test target effect and content preservation for continuous outputs such as categorical and severity targets in medical diffusion. Abductive counterfactual [Shen et al. 2025] inference is used in image-quality assessment to explore deep features and perceptual distortion links. Regression counterfactuals can reveal false monotonic relationships, but a smooth change in model output doesn't mean the visual factor varies monotonically in reality. Evaluation should measure target error at multiple magnitudes and ensure invariant content does not drift.

The world-validity ceiling. General classification is useful for evaluating search algorithms and generators, but is weak for asserting that the resulting images depict achievable reality. Perturbation actions used to transform images are

typically underspecified; individuals cannot perform pixel edits, and changes to object classes do not guarantee a feasible intervention. Even semantic reliability checks [Cao et al. 2026], which are key to identifying correctly transformed images, mostly assess agreement with evaluators rather than causal attainability. Thus, general classification mainly tests contrast, preservation, and model dependence, supporting causal language only when external actions are modeled.

7.2 Medical and Biomedical Imaging

Medical imaging underscores the clinical importance of the feature-preservation challenge. A counterfactual that appears visually realistic may still be anatomically nonsensical, inconsistent with the same patient's data, incompatible with disease progression, or potentially misleading for its intended audience.

Case-based explanation, identity, and privacy. Medical case explanations are intuitive because clinicians often reason by comparison, but showing a retrieved or generated case creates privacy and identity risks. Montenegro et al. jointly optimize realism, privacy proxies, and explanatory evidence for factual and counterfactual case images [Montenegro et al. 2021]. Their follow-up work on identity-disentanglement study continues this direction [Montenegro and Cardoso 2024]. Valoor et al. use case-based counterfactual maps in Alzheimer diagnosis from structural MRI [Valoor and Gangadharan 2025]. These systems present two primary concerns regarding preservation: whether the diagnostic morphology is preserved and whether identifying information is adequately concealed. Robust anonymization techniques may potentially remove subtle pathological features, whereas an image that remains clinically accurate can serve to maintain biometric identity.

Morphology and patient-level semantics. CounterSynth [Pombo et al. 2023] generates morphologically constrained 3D brain images under counterfactual demographic conditions and evaluates their use in equitable model training. Singla et al. condition generated paths on diagnostic probability and audit changes using radiographic measures and radiology assessment [Singla et al. 2023a]. GANterfactual [Mertes et al. 2022] translates between abnormal and normal chest-image domains and evaluates whether the resulting contrasts help medical non-experts understand a classifier.

Disease severity and structured regions. Diffusion-autoencoder traversal extends medical counterfactuals from binary classification to multi-level and continuous targets across several imaging tasks [Atad et al. 2024]. MiMICRI [Guo et al. 2024] takes a complementary, domain-centered approach by allowing cardiovascular experts to exchange segmented morphological regions and inspect the classifier response.

Discovery, auditing, and weak supervision. Some medical counterfactual systems are better viewed as *instruments for hypothesis generation* than as patient-specific explanations. Lang et al. discover visual factors across retinal, external-eye, and chest-radiograph models and submit the resulting hypotheses to interdisciplinary review [Lang et al. 2024]. DISCOVER [Rotem et al. 2024] identifies and exaggerates latent factors associated with embryo-quality classification, comparing them with known morphological properties and supporting instance-level analysis. Counterfactual relations also provide weak or contrastive supervision for lesion segmentation, whole-slide classification, and radiology-report generation [Li et al. 2024b; Lin et al. 2024; Shvetsov et al. 2024]. Medical VQA intervention [Cai et al. 2024], on the other hand, masks visually relevant regions to test and improve image-answer.

The medical-imaging literature delineates a set of validity requirements: (i) *Image realism:* does the output closely resemble an image from the modality? (ii) *Anatomical coherence:* are structures, topology, scale, and acquisition characteristics mutually consistent? (iii) *Patient preservation:* does the alternative retain the individual characteristics declared invariant? (iv) *Clinical plausibility:* is the proposed change compatible with pathology, covariates, disease

stage, and the clinical question? (v) *Clinical usefulness:* does the artifact enhance an appropriate user's interpretation or decision without inducing harmful confidence?

7.3 RL, Autonomous Driving, and Embodied Visual Systems

Visual counterfactuals for sequential systems replace the classifier's label with an action, policy decision, or planned consequence. The pertinent counterfactual, therefore, extends beyond merely considering "what image alters the action?" to asking under which attainable history or environmental intervention the policy would respond differently, and what follow-up outcomes might subsequently ensue?

From saliency hypotheses to state alternatives. Atrey et al. use counterfactual changes in Atari environments to test behavioral hypotheses derived from saliency maps [Atrey et al. 2020]. Olson et al. generate visual states that induce another agent action and show them to non-expert participants [Olson et al. 2021]. GANterfactual-RL [Huber et al. 2023] extends generated alternatives toward comparing learned strategies rather than isolated actions. Counterfactual World Modeling probes physical dynamics through alternative masking conditions, Point Prompting uses counterfactual video regeneration as a motion-tracking interface, and CREStE uses counterfactual trajectory guidance in mapless navigation [Shrivastava et al. 2026; Venkatesh et al. 2024; Zhang et al. 2025a]. These contributions move evaluation beyond image quality toward behavior, dynamics, or downstream control. SAFE-RL [Samadi et al. 2024] explicitly accounts for a history of visual observations when saliency-guided counterfactuals are generated for deep RL policies.

Driving scenes and object semantics. Autonomous driving adds structured scenes, multiple agents, sensor geometry, traffic rules, and asymmetric failure costs. OCTET [Zemni et al. 2023] represents a scene through objects and permits controlled spatial or stylistic edits, making "what changed" more intelligible than unrestricted latent movement. SAFE [Samadi et al. 2023] uses saliency to focus counterfactual changes for driving decisions, while semantic Shapley counterfactuals [Sun et al. 2025] estimate influential scene elements and generate targeted alternatives. Adaptive masked diffusion [Guo et al. 2026] is also evaluated on driving imagery, demonstrating that localized generation can operate beyond faces and object-centered datasets.

For driving, at least five types of intervention should be distinguished: (i) an *appearance intervention*, such as colour, texture, illumination, or weather; (ii) a *sensor intervention*, such as blur, occlusion, exposure, or camera viewpoint; (iii) a *scene intervention*, such as adding, removing, or relocating an object; (iv) an *agent intervention*, such as changing another road user's intent or trajectory; and (v) an *ego-action intervention*, such as braking, yielding, or changing lane.

7.4 VQA, VLMs, and Multimodal Reasoning

In multimodal systems, the factual-counterfactual relation can be placed in the image, question & answer, causal graph, or an internal joint representation. This flexibility makes the word "counterfactual" particularly easy to overextend.

Counterfactual supervision for VQA.. VQA models often exploit language priors rather than the image evidence needed for the question. Bi et al. use counterfactual samples and self-supervised knowledge distillation to preserve relations between factual and altered examples, thereby improving robustness on VQA benchmarks [Bi et al. 2024]. CCIS-MVQA [Cai et al. 2024] estimates the effect of relevance-guided visual masking and uses it during medical VQA training. Generated or edited image–text pairs further support compositional concept learning, set-based CLIP training, and counterfactual referring-expression comprehension [Jia et al. 2025; Lai et al. 2024; Yu and Li 2024].

Structured reasoning and counterfactual ground truth. Video questions require reasoning over event order and dependence rather than one static object. CounterVQA [Chen et al. 2026] represents causal relations through generated graphs and uses them to construct progressively difficult factual and counterfactual questions; graph-guided post-training then improves performance within this regime. CHARTOGRAPHER [Jiang et al. 2026] reconstructs charts as executable programs, changes the underlying data, rerenders the chart, and recomputes the answer. VLM Reality Check [Sar et al. 2026] applies controlled minimal transformations across factors such as texture, background, object presence, orientation, and visual-typographic conflict. These benchmarks test whether model behavior follows the changed visual fact rather than a fixed question template or learned prior.

Hallucination diagnosis and mitigation. HalCECE [Lymperaiou et al. 2025] uses conceptual and scene-graph counterfactuals to distinguish missing, additional, and relational errors in image captions. Antidote [Wu et al. 2025] uses counterfactual supervision and preference optimization to mitigate false presuppositions and object hallucination, whereas Envision–Attend–Respond [Liang et al. 2026] constructs a diffusion-derived visual impression, compares it with the original through the VLM's visual pathway, and suppresses tokens associated with conflict.

These studies expose three different notions of multimodal faithfulness. *Input faithfulness* evaluates if the counterfactual edit affects only the targeted visual element. *Cross-modal faithfulness* assesses whether the answer or caption adjusts in response to that visual change. *Mechanistic faithfulness* investigates whether the artifact identifies an internal cause behind the original output. Counterfactual pairs effectively test the first two types but do not necessarily satisfy the third. Claims of better grounding or detected inconsistencies should not be overinterpreted as explanations of the internal reasoning of VLMs unless their internal evidence is independently validated.

7.5 Anomaly Detection, Repair, and Weak Localization

Visual anomaly tasks invert the usual counterfactual target. In class explanation, the foil can be a named category; in anomaly explanation, the target is typically "normal." Normality is a distribution; there may be many ways to reduce an anomaly score, only some of which correspond to the defect, pathology, or process failure of interest. This underdetermination makes constructing a normal counterpart central to the explanation.

AR-Pro [Ji et al. 2024] defines a counterfactual as a diffusion-based repair and formalizes properties relative to a linearly decomposable anomaly detector. COIN [Shvetsov et al. 2024] generates an abnormal-to-normal medical image through inpainting and uses the residual as a weak segmentation mask.

The cross-application synthesis suggests that, for image-generating methodologies, visual plausibility is vital but insufficient. Each domain incorporates an intervention model that depicts what can be perturbed and what must remain unchanged. Progress should be assessed not solely by enhancements in generators or flip rates but also by the clarity with which each method articulates and examines its intervention assumptions.

8 Evaluation Metrics

Evaluation of visual counterfactuals must distinguish prediction change, perceptual preservation, semantic correctness, causal consistency, and practical usefulness. These properties are related but not interchangeable: a class flip changes model behavior, not the realism of the edited image; visual similarity does not establish that the intended intervention occurred; and distributional realism does not demonstrate individual-level causal validity. Evaluation must also reflect the counterfactual's role. Explanatory methods require evidence about model behavior and, where claimed, user understanding, whereas counterfactual learning, auditing, and repair require corresponding downstream outcomes.

Table 4. Application-specific semantics and validation requirements for visual counterfactuals.

Application	Counterfactual target	Domain-specific invariants	Evidence needed beyond a target change
General/fine-grained classification	Alternative class, attribute, similarity judgment, or continuous output	Instance content, identity, background, pose, and target-irrelevant semantics	Independent semantic correctness, matched comparison conditions, and an action or process model for world-valid claims
Medical imaging	Diagnosis, severity, morphology, case, or localized pathology	Anatomy, patient identity, acquisition conditions, covariates, and clinical context	Anatomical and clinical validation, privacy/patient preservation, external-site testing, and recipient-specific usefulness
RL and autonomous systems	Alternative state, action, policy response, trajectory, or plan	Reachable history, dynamics, scene geometry, agent interaction, and safety constraints	Environment-valid intervention, rollout consequences, scenario coverage, and human or operational safety evidence
VQA and VLMs	Alternative answer, caption relation, event outcome, or grounding response	Unchanged question intent, protected visual factors, cross-modal alignment, and recomputable truth	Pair verification, expected response relation, leakage controls, and separation of grounding from mechanistic claims
Anomaly and repair	Normal counterpart, localized defect, or repaired instance	Non-anomalous content, object/process structure, and feasible repair conditions	Detector-independent localization, normality coverage, domain feasibility, and preservation of safety- or diagnosis-relevant evidence

Table 5 summarises the principal metrics and tests, including specialized VCE measures and the axiomatic framework of Monteiro et al. [Monteiro et al. 2023].

Meaningful comparison requires a declared target or foil, an explicit set of preserved properties, and a reproducible evaluation protocol. Studies should specify preprocessing, evaluation models, thresholds, sample sizes, random seeds, and whether scores are computed over all attempted generations or only successful counterfactuals. Success-conditioned quality scores should be accompanied by the overall success rate and failure count. Where possible, semantic verification should be independent of the target model and generator, with evaluator uncertainty acknowledged. Because proximity, validity, diversity, and realism can conflict, evaluation should expose trade-offs rather than collapse them into an unexplained aggregate score. Axiomatic consistency provides additional necessary checks for causal image models, but does not by itself identify the true counterfactual outcome.

9 Synthesis Summary and Open Challenges

9.1 The Key Takeaway

Following this review, a recurring theme is that visual counterfactuals can produce measurable benefits; however, these benefits are specific to the role, task, and validation protocol under which they were demonstrated.

When counterfactuals are designed with humans as the intended users, the studies show that counterfactual and alterfactual contrasts can communicate distinctions tested by the model or help users identify a flawed agent [Mertes et al. 2024; Olson et al. 2021; Zhao et al. 2021]. On the other hand, system-level studies show that counterfactual relations have supported compositional and procedural representation learning [Kung et al. 2025; Lai et al. 2024], radiology and pathology supervision [Li et al. 2024b; Lin et al. 2024], controlled evaluation of hypothetical reasoning [Ness et al. 2026; Yu and Li 2024; Zhang et al. 2024b], and mitigation of selected multimodal failures [Howard et al. 2024; Zong et al. 2025a]. These findings demonstrate utility as teaching material, training data, diagnostic evidence, or system interventions.

Table 5. Evaluation metrics and tests for visual counterfactuals. Arrows indicate preferred directions, conditional on achieving the intended intervention. Metrics assess complementary properties rather than a universal notion of counterfactual quality.

Metric or test	Measurement	Interpretation and limitations
Flip rate / target success ↑	Proportion of attempts changing the predicted class or reaching a specified foil [Jeanneret et al. 2023].	Distinguish untargeted flips from targeted success. Neither establishes meaningful edits; non-classification tasks require explicit success rules.
COUnterfactual Transition (COUT) ↑	Target-minus-original class area under perturbation curves during ordered insertion of counterfactual pixels [Khorram and Fuxin 2022].	Rewards early prediction transitions; it depends on pixel ordering and step size. Intermediate images may be off-manifold.
Pixel distance / edit sparsity ↓	ℓ_1, ℓ_2, ℓ_∞, ℓ_0, thresholded edit area, or structured edit counts.	Representation-, resolution-, and threshold-dependent; small changes need not be meaningful or globally minimal.
SSIM / PSNR ↑; LPIPS ↓	Structural similarity, reconstruction fidelity, and learned perceptual distance [Wang et al. 2004; Zhang et al. 2018].	Resemblance does not establish the correctness of the intervention; necessary semantic changes may reduce similarity.
Face Verification Accuracy (FVA) / Face Similarity (FS)	Same-identity verification rate and face-embedding similarity between factual and counterfactual images [Jeanneret et al. 2023].	Face-specific and evaluator-dependent. FVA and cosine similarity favor higher values; cosine distance favors lower values.
SimSiam Similarity (S^3) ↑	Cosine similarity between SimSiam embeddings of image pairs [Jeanneret et al. 2023].	Measures representation-level preservation rather than explicit semantic correctness; relevant changes may be missed.
Mean Number of Attributes Changed (MNAC) ↓	Mean count of changed attribute predictions [Jeanneret et al. 2022; Rodríguez et al. 2021].	Declare vocabulary and target inclusion. Low counts can penalize necessary dependent edits or hide spurious associations.
Correlation Difference (CD) ↓	$\sum_a \lvert c_{qa} - c_{qa}^{\mathrm{CF}} \rvert$: difference between reference attribute correlations and correlations of signed attribute changes [Jeanneret et al. 2022].	Dataset- and oracle-dependent. Matching observational associations does not establish causal validity or internal model mechanisms.
Semantic / invariant checks	Attribute agreement, relation satisfaction, mask overlap, keypoint error, or expert assessment.	Verify intended changes separately from preserved properties; account for evaluator error and shared biases.
FID / KID ↓	Feature-space discrepancy between generated and reference image distributions [Heusel et al. 2017; Mikołaj et al. 2018].	Population-level realism, not pairwise correctness, is affected by reference data, sample size, and feature extractor.
Split-FID (sFID) ↓	FID against a disjoint factual subset, averaged over repeated splits [Jeanneret et al. 2023].	Reduces source-image overlap bias but retains FID's limitations. Here, SFID means split-FID, not spatial FID.
Diversity/coverage	Perceptual or semantic spread, distinct edit sets, and coverage of valid alternatives.	Separate explanatory alternatives from nuisance variation; report validity, sampling budget, rejection, and duplicates.
Composition error ↓	Deviation from the original image under null interventions, including repeated applications [Monteiro et al. 2023].	Tests factual preservation. An identity-copying model can pass, so effectiveness must also be assessed.
Reversibility error ↓	Reconstruction error after an intervention followed by restoration of factual parent values [Monteiro et al. 2023].	Tests round-trip consistency, not correctness of the intermediate counterfactual or causal identification.
Effectiveness ↑	Pseudo-oracle agreement with requested parent values, using suitable classification or regression measures [Monteiro et al. 2023].	Tests intervention realization, not merely classifier flipping; depends on oracle reliability under edited combinations.
Faithfulness/robustness tests	Controlled edits, ablations, internal interventions, and stability across justified perturbations or seeds.	Input–output sensitivity alone does not identify mechanisms; legitimate alternative explanations need not be identical.
Human usefulness	Model-simulation accuracy, error detection, decision quality, appropriate reliance, and completion time.	Use input-only and prediction-only baselines. Preference or increased trust alone does not demonstrate understanding.
Downstream / domain outcomes	Accuracy, calibration, fairness, robustness, hallucination, segmentation, repair, or safety performance.	Use held-out, resource-matched comparisons. Task gains do not independently establish counterfactual correctness.
Computational efficiency ↓	Latency, memory, model queries, optimization iterations, and sampling steps.	Separate training from inference; include failures and retries, and match hardware, resolution, quality, and success.

Furthermore, the common contribution across these roles is the use of an explicit contrast. A specified foil, alternative action, or expected answer relation makes the tested factual instance more precise than an undirected statement of importance [Kung et al. 2025; Zhang et al. 2025b]. The reviewed papers distinguish perceptual or manifold quality [Augustin et al. 2024; Zaher et al. 2026], semantic reliability [Cao et al. 2026], and classifier-relative or representational faithfulness [Chowdhury et al. 2025; Haselhoff et al. 2024]. Negative findings reinforce that counterfactual analysis can falsify a saliency-based explanation [Atrey et al. 2020], while expert interaction can reveal physiologically implausible edits [Guo et al. 2024], and usefulness can vary across visual concepts [Law et al. 2023]. The primary challenge, therefore, is establishing the evidential chain that links a declared intervention to independent verification and to the outcome asserted for its intended purpose.

9.2 Open Challenges and Research Directions

The reviewed literature establishes visual counterfactuals as more than an explanation; they also provide structured contrasts for representation learning, model auditing, benchmark construction, and inference-time intervention. Across these roles, generating an alternative image or response is only one component of the problem. More fundamental questions concern which variables can be intervened upon, which relationships must be preserved, what makes a counterfactual attainable, and what can legitimately be inferred from the resulting contrast. This section develops a connected research agenda around these unresolved questions. Table 6 summarizes the open questions and links their motivation to earlier sections.

A fundamental challenge involves determining who defines the semantic intervention vocabulary, whether its variables are accurately represented in pixels, and whether they capture distinctions recognized by both humans and models [Spanos et al. 2026]. Relational validity additionally necessitates distinguishing between intended alterations, essential consequences, and preserved invariants. Overcoming the limitations of world-validity requires supplementary grounding through physical observations or executable mechanisms, rather than relying on realism alone [Ness et al. 2026; Winter et al. 2024]. In situations where alternatives are underdetermined, methods should reflect uncertainty or choose to abstain. This is particularly significant in the context of anomaly repair, where different acceptable "normal" counterparts may lead to divergent explanations [Ji et al. 2024; Shvetsov et al. 2024].

Further research should investigate the circumstances under which visual edits correspond to internal concepts or activation interventions, thereby extending the existing body of work on model-faithful explanations [Khan et al. 2024]. Foundation models introduce additional challenges when they generate, judge, and respond to counterfactuals: open-ended foils necessitate explicit semantics, and agentic auditing must distinguish between competing hypotheses through confirmed evidence held out for validation [Rott Shaham et al. 2024]. Grounded testbeds and shared protocols can facilitate these investigations; however, benchmark-guided post-training evaluation also encourages examining whether learned capabilities are transferable beyond familiar construction rules [Chen et al. 2026].

Finally, conflicting objectives necessitate selecting context-dependent solutions rather than a universally optimal explanation. Probabilistic and Pareto-based formulations explicitly acknowledge these trade-offs without determining a suitable balance for every user [Del Ser et al. 2024; Vaeth et al. 2025]. Interactive preference elicitation can reveal alternatives while maintaining essential constraints. Privacy-oriented counterfactual approaches further encourage broader examination of issues such as data leakage, selective disclosure, bias introduced by generators, and potential manipulation [Montenegro and Cardoso 2024]. Collectively, these priorities shift the focus from persuading with alternatives toward evidence based on counterfactuals, whose significance, mechanisms, limitations, and intended applications are justifiably defended.

Table 6. Summary of Open Research Questions

Research direction	Motivation	Open questions
Semantic vocabularies and human–model alignment	Section 5.2 shows that attribute, prototype, and latent-factor methods inherit different semantic priors. Section 8 highlights the importance of domain interpretation, illustrated by three expert-engagement examples [Guo et al. 2024; Lang et al. 2024; Rotem et al. 2024].	How can grounded, revisable, multilevel edit vocabularies be developed? When do concept-space edits translate faithfully into pixels, and do humans and models respond to the same semantic changes?
Relational validity and compositional interventions	Sections 5.5–5.6 and 7.4 show that object removal, event modification, and chart editing require different dependent changes. ObjectDrop, CounterVQA, and CHARTOGRAPHER illustrate complementary ways to constrain these relationships [Chen et al. 2026; Jiang et al. 2026; Winter et al. 2024].	How can methods distinguish manipulated variables, permitted consequences, and invariants? Under what conditions can interventions be composed without introducing order-dependent or cross-modal contradictions?
World-validity ceiling	Sections 5.5 and 7.1–7.3 distinguish realistic depictions from attainable alternatives. Morphological constraints, physical observations, and executable worlds provide bounded grounding rather than universal feasibility [Ness et al. 2026; Winter et al. 2024].	What action models, transition rules, or additional observations support world-valid counterfactuals? When should methods return multiple admissible alternatives, request further evidence, or abstain?
Input-space and representation-space correspondence	Sections 6.2, 7.4, and 8 distinguish behavioral sensitivity from internal-mechanism identification. Concept interventions and contribution-based methods provide partial evidence, but do not establish general correspondence between visual edits and internal computations [Khan et al. 2024; Li et al. 2024a].	When do visual edits agree with activation patching, concept steering, or sparse-autoencoder interventions? How can agreement and disagreement distinguish competing mechanisms rather than merely corroborate a preferred explanation?
Foundation models and agentic auditing	The access dependencies introduced in Section 4 become consequential in Sections 5.6, 6.2, 6.4, and 7.4, where learned components support multimodal diagnosis and mitigation [Liang et al. 2026; Lymperaiou et al. 2025; Wu et al. 2025].	How should foils be defined for open-ended outputs? How can editors and judges be assessed without complete ground truth, and how can auditing agents separate adaptive exploration from held-out hypothesis confirmation?
Grounded testbeds and benchmark-independent learning	Section 6.3 discusses six controlled-evaluation exemplars. Together with the learning applications in Sections 6.1 and 7.4, these expose the risk of learning construction regularities rather than transferable intervention reasoning.	How can complementary testbeds and a shared intervention protocol support reproducible research? Which learning strategies transfer across intervention compositions, renderers, and dependency structures without instance leakage or construction-process overfitting?
Underdetermined normality in anomaly repair	Sections 6.4 and 7.5 develop the repair problem through AR-Pro and COIN [Ji et al. 2024; Shvetsov et al. 2024], illustrating why reducing an anomaly score does not uniquely identify the defect, its appropriate replacement, or a feasible repair.	How can context-conditioned, potentially multimodal normality be represented? Which conclusions remain stable across admissible normal counterparts, and how can reconstruction uncertainty be separated from anomaly evidence?
Conflicting objectives and preference-sensitive trade-offs	Sections 4, 5.2, and 8 reveal tensions among output scope, preservation, target effectiveness, and usefulness. Pareto-based generation makes objective conflicts explicit but leaves the selection of an appropriate alternative unresolved [Del Ser et al. 2024].	Which properties are mandatory constraints and which are negotiable? Can preference elicitation and interactive trade-off exploration improve understanding and decision quality without encouraging selective narratives or excessive cognitive burden?
Security, privacy, and manipulation	Sections 6.4 and 7.2 identify two privacy-oriented studies concerning generated case explanations and identity disentanglement [Montenegro and Cardoso 2024; Montenegro et al. 2021]. These motivate privacy–utility questions without establishing the prevalence of broader security threats.	How do counterfactual interfaces affect extraction, membership inference, leakage, and selective-disclosure risks? How can provenance and adversarial testing distinguish generator-induced bias from target-model behavior and resist manipulation of explanatory evidence?

10 Conclusion

This review provides a comprehensive overview of visual counterfactual research, including recent targeted applications. From the review, we understood that visual counterfactuals are best regarded as evidence-bearing interventions that connect a factual instance to a specified alternative. This indicates that the field has progressed beyond merely flipping a particular class of images or serving solely as explanation algorithms. The interpretation of visual counterfactual algorithms depends on the target system, the contrast under analysis, the permissible modifications introduced by the intervention, and the properties that must remain invariant. This perspective addresses a longstanding ambiguity in the literature: the same artifact may serve multiple purposes, such as explanation, learning, auditing, evaluation, or system modification, yet success in one function does not necessarily validate claims in another.

The reviewed literature shows that more advanced construction mechanisms have increased the realism, scale, and scope of visual counterfactuals. However, the quality of artifacts and their evidential validity remain separate issues. A change in prediction alone does not guarantee semantic accuracy; perceptual plausibility does not confirm domain feasibility; and agreement with another trained model does not necessarily mean independent verification. Consequently, the credibility of a counterfactual claim relies on a clear chain of evidence linking the intended intervention to the preservation of key features, model or domain fidelity, and the outcome's relevance to its technical purpose.

The field should consequently transition from method-centered comparison to claim-centered evaluation. Progress is demonstrated when a study clearly specifies its contrast and permissible intervention, discloses its access and auxiliary-model assumptions, assesses relevant failure modes, and confines its conclusions to the evidence supported. By separating artifact from role and aligning evaluation with the asserted claim, the framework developed in this review facilitates meaningful comparison without requiring the imposition of a universal scale across heterogeneous methods. In accordance with this standard, visual counterfactuals can offer more than merely plausible alternative images; they can serve as flexible instruments for examining model behavior, testing visual dependencies, and intervening in learning and decision-making processes. Consequently, their long-term value will hinge not only on whether alternatives become indistinguishable from observations, but also on whether their meaning, validity, and implications can be independently scrutinized and reproduced.

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